

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Assignment 03



Submitted to:

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Answer 1

A repeater regenerates and amplifies digital signals to restore their original shape and strength. It removes noise before retransmitting the signal, ensuring better quality while an amplifier simply increases the amplitude of both the signal and any noise present, without differentiating between them. This can result in a distorted signal if noise levels are high.

Answer 2

A switch can decide whether to forward or block data packets based on MAC addresses and the intended destination. This process is called filtering.

Importance:

- Prevents unnecessary traffic from reaching unrelated devices.
- Reduces network congestion.
- Enhances security by ensuring data packets reach only the intended recipient.

Answer 3

A hub is essentially a multi-port repeater. It amplifies and retransmits incoming signals to all connected devices on its network. Like a repeater, it does not filter or direct traffic, leading to potential collisions in the network.

Answer 4

Forwarding Port actively forwards network traffic to its destination. It is part of the active network path whereas Blocking Port prevents data from being forwarded, usually to avoid loops in the network (as in STP - Spanning Tree Protocol). It listens to network traffic but does not forward it.

Answer 5

Time: VLANs simplify network management by grouping devices logically, regardless of their physical location. This reduces the complexity of reconfiguring hardware when changes are needed.

Money: By reducing the need for additional physical hardware, like routers, VLANs optimize resource usage and cut down infrastructure costs.

Answer 6

VLANs segment a network into separate, isolated broadcast domains. This limits communication between VLANs unless explicitly allowed, thereby:

- Preventing unauthorized access to sensitive data.
- Containing security breaches within a single VLAN.
- Reducing attack surfaces for malicious users.

Answer 7

a) **110.11.5.88**

Ans. **Binary:** 01101110.00001011.00000101.01011000

b) **12.74.16.18**

Ans. **Binary:** 00001100.01001010.00010000.00010010

c) **201.24.44.32**

Ans. **Binary:** 11001001.00011000.00101100.0010000

Answer 8

a) **01011110 10110000 01110101 00010101**

Ans. **Decimal:** 94.176.117.21

b) **10001001 10001110 11010000 00110001**

Ans. **Decimal:** 137.142.208.49

c) **01010111 10000100 00110111 00001111**

Ans. **Decimal:** 87.132.55.15

Answer 9

Concept of Hop Count in RIP (Routing Information Protocol): RIP uses hop count as a metric to measure the distance between the source and destination in a network. Each hop refers to a transition from one router to the next in the path. The hop count value starts at 0 and increases by 1 for every router a packet passes through. RIP limits the maximum hop count to 15, with 16 indicating an unreachable destination.

The reason no hop counted between N1 and R1 is that in Figure 1, N1 (source) is directly connected to R1 via a local network (typically a LAN connection). The connection between a host (N1) and the first router (R1) is not counted as a "hop" because RIP only counts hops between routers, not between end devices and the router. The first hop is counted only when a packet moves from R1 to the next router (R2).

Answer 10

To find the shortest distance between node x and node y using the Bellman-Ford equation, we can follow the principle of relaxation. Here are the given values for the distances and costs:

Day = 5, Dby = 6, Dcy = 4, Ddy = 3

Cxa = 2, Cxb = 1, Cxc = 3 Cxd = 1

To calculate Dxy, we use the Bellman-Ford equation:

$D_{xy} = \min(D_{ay} + C_{xa}, D_{by} + C_{xb}, D_{cy} + C_{xc}, D_{dy} + C_{xd})$

$D_{xy} = \min(5+2, 6+1, 4+3, 3+1)$

$D_{xy} = 4$

Therefore, the shortest distance between node x and node y is 4